Project Design Phase-II

Data Flow Diagram & User Stories

Date	08 March 2026
Team ID	—
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis
Maximum Marks	4 Marks

Data Flow Diagram – Level 0 (Context Diagram)

The system receives health parameters from the user, processes them through the Flask web server and machine learning model, predicts the hypertension stage, and returns the prediction result along with medical recommendations to the user.

Entity / Process	Data Flow	Description
User → Predictive Pulse System	Health Parameters Input	User enters health information such as age group, symptoms, blood pressure levels, medication history, and diet control through the web interface
Web Interface → Flask Server	Form Submission	HTML form sends user input data to Flask backend through HTTP POST request
Flask Server → Data Processing Module	Data Preprocessing	Input data is validated, encoded, and scaled to match the format required by the machine learning model
Processing Module → ML Model	Prediction Request	Preprocessed data is passed to the Logistic Regression model for hypertension stage prediction
ML Model → Flask Server	Prediction Result	Model returns predicted hypertension stage and probability score
Flask Server → Recommendation Engine	Health Advice Generation	System selects appropriate medical recommendations and lifestyle suggestions based on predicted stage
Predictive Pulse System → User	Prediction Output	Results including hypertension stage, risk level color indicator, confidence score, and health recommendations are displayed on the web interface

User Stories

USN	User Type	Epic	User Story / Task	Acceptance Criteria	Priority	Sprint
USN-1	End User	Health Input	As a user, I can enter my health parameters in the form to check hypertension risk	User inputs accepted and validated successfully	High	Sprint-1
USN-2	End User	Risk Prediction	As a user, I can receive prediction results indicating hypertension stage	System displays predicted stage and risk level	High	Sprint-1
USN-3	End User	Health Recommendations	As a user, I can receive medical advice and lifestyle recommendations based on prediction	System shows relevant recommendations	High	Sprint-2
USN-4	End User	Risk Visualization	As a user, I can view risk level with color indicators and confidence score	Results displayed clearly on interface	Medium	Sprint-2
USN-5	Developer	Model Training	As a developer, I can train the machine learning model	Model successfully trained and saved	High	Sprint-1

			using the hypertension dataset			
USN-6	Developer	Backend System	As a developer, I can integrate the machine learning model with Flask server	Model predictions accessible through backend	High	Sprint-1
USN-7	Developer	Web Interface	As a developer, I can build a web interface for user interaction	HTML form connects with backend successfully	High	Sprint-1
USN-8	Developer	System Integration	As a developer, I can connect frontend, backend, and machine learning model	End-to-end prediction works correctly	High	Sprint-2
USN-9	Developer	Testing	As a developer, I can test the system using various health inputs	System produces accurate predictions without errors	Medium	Sprint-2